

MSS (2004) Revisited: Weak Instruments and Dynamic Panel Identification

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1 Introduction

Estimating the causal effect of income shocks on civil conflict is not an estimation problem but an identification problem. OLS confounds three biases: conflict destroys output (reverse causation), national accounts in low-income economies are measured with error, and persistent country traits affect both (omitted variables). No amount of data fixes this. The effect is identified only if we can find a source of variation in growth that is exogenous to conflict and does not enter the conflict equation directly.

Miguel et al. (2004), henceforth MSS, proposed that source: annual rainfall growth, used as an excluded instrument for per-capita GDP growth on a panel of 41 sub-Saharan African countries over 1981–1999. The identifying assumptions are that rainfall shifts agricultural output (relevance), and that it affects conflict only through that channel (exclusion). The paper was novel in 2004. It also predates several of the diagnostic tools we now use to evaluate IV designs.

The literature that followed targeted each of these assumptions in turn. Brückner and Ciccone (2010) proposed an alternative instrument based on global commodity prices, relying on a different exogeneity argument. Ciccone (2011) questioned the specification of the first stage, showing that rainfall’s mean-reversion contaminates the growth-as-IV interpretation. Sarsons (2015) tested the exclusion restriction directly, exploiting irrigation infrastructure that breaks the agricultural channel.

We revisit MSS with two questions. First, how much the original rainfall design tells us about β once modern weak-instrument diagnostics are applied. Second, whether a dynamic-panel identification strategy that does not rely on rainfall recovers a coherent estimate.

Section 2 formalises the MSS identification. Section 3 surveys the critiques. Section 4 describes the data. Section 5 sets out the estimators. Section 6 reports estimates. Section 7 concludes.

2 The MSS identification strategy

MSS fit the structural equation

$$C_{it} = \alpha_i + \lambda_i t + \beta g_{it}^y + \gamma' X_{it} + u_{it}, \quad (1)$$

where $C_{it} = \mathbf{1}\{\text{conflict}_{it}\}$ is a binary indicator for civil conflict in country i at year t , $g_{it}^y = \Delta \log(\text{GDP}_{it})$ is per-capita GDP growth, α_i are country fixed effects, $\lambda_i t$ are country-specific linear time trends, and X_{it} collects time-varying controls (lagged Polity 2). Opportunity-cost theory (Grossman, 1991) predicts $\beta < 0$.

Estimating equation (1) by OLS is inconsistent for the reasons listed in Section 1. MSS instrument g_{it}^y with annual rainfall growth g_{it}^R and its first lag:

$$g_{it}^y = a_i + b_i t + \pi_1 g_{it}^R + \pi_2 g_{i,t-1}^R + \delta' X_{it} + v_{it}. \quad (2)$$

The identifying assumptions, made conditional on α_i and X_{it} , are:

Relevance: $\text{Cov}(g_{it}^R, g_{it}^y) \neq 0$. Rainfall shifts agricultural output, which is a large share of African GDP. Testable through the first-stage F statistic on the excluded instruments.

Exclusion: $\text{Cov}(g_{it}^R, u_{it}) = 0$. Rainfall affects conflict only through GDP. Not directly testable in general; with two instruments, the Sargan test provides one degree of freedom of indirect check on whether the two rainfall measures imply the same β .

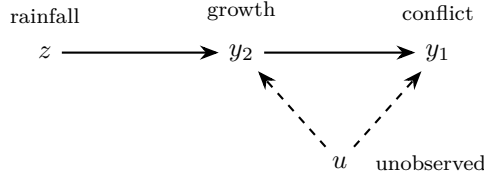


Figure 1: The MSS identification strategy as a DAG. Solid arrows show the assumed causal pathway: rainfall (z) affects growth (y_2), which affects conflict (y_1). Dashed arrows show unobserved confounders (u) that bias OLS. Identification of β requires the $z \rightarrow y_2$ arrow (relevance, testable) and the *absence* of any $z \rightarrow y_1$ arrow (exclusion, not directly testable).

The specification choices matter. MSS use country fixed effects to absorb time-invariant heterogeneity (geography, institutions, colonial history) and country-specific linear time trends to absorb slow secular movements (democratisation, technology diffusion). They do not use year fixed effects, which would absorb the common component of rainfall shocks across the panel and reduce the variation the instrument exploits. Standard errors are clustered at the country level.

Their preferred 2SLS specification yields $\hat{\beta} \approx -2.4$: a five-percentage-point decline in growth raises the probability of conflict by roughly twelve percentage points.

3 Methodological critiques and alternative identification

The literature that followed MSS extended the design along three axes: it offered alternative instruments, questioned the first-stage specification, and tested the exclusion restriction directly. Only the last two are critiques in the strict sense; the first is a parallel identification strategy.

Alternative instrument. Brückner and Ciccone (2010) replace rainfall with country-specific commodity export price shocks: the weighted sum of global commodity prices, using each country's export composition as weights. The exogeneity argument is small-country price-taking. Global commodity prices are determined on world markets, and no single sub-Saharan African

country is large enough to move them. Their first stage becomes

$$g_{it}^y = a_i + \tilde{\pi} \text{shock}_{it}^P + \tilde{\delta}' X_{it} + v_{it}, \quad (3)$$

while the structural equation (1) is unchanged. They recover a negative and significant β , corroborating the MSS sign under a different identifying assumption. This is identification pluralism rather than critique: two unrelated exogeneity arguments converge on the same conclusion, which strengthens MSS rather than weakens it. Within MSS's own design, the analogue is the Sargan over-identification test on the two rainfall instruments (current and lagged); we return to it in Section 6.

Specification critique. Ciccone (2011) attacks the functional form rather than the exogeneity. Rainfall is mean-reverting: an unusually wet year tends to be followed by drier years, so rainfall growth carries a mechanical negative serial correlation with the past level of rainfall. If conflict responds to rainfall *levels* (drought leads to a bad harvest, which raises conflict), then instrumenting growth with rainfall growth captures a level-driven effect dressed up as a growth effect. He re-runs the analysis with rainfall levels as the IV and finds the effect attenuates substantially. Miguel and Satyanath (2011) reply defending the original parameterisation. The lesson is generic: 2SLS consistency requires correct first-stage specification, not just instrument exogeneity. An exogenous instrument applied in the wrong functional form still produces an inconsistent estimate.

Direct test of exclusion. The most consequential critique tests the exclusion restriction itself. Sarsons (2015) uses Indian rainfall and conflict data combined with records of irrigation dams. In regions *downstream of dams*, water supply is decoupled from local rainfall: the agricultural-income channel is physically broken. If MSS's exclusion holds (rainfall affects conflict only through agricultural income), then in dam-downstream regions rainfall should not predict conflict. She finds that it does. The implication is a direct $z \rightarrow y_1$ arrow that the MSS DAG denies (Figure 2).

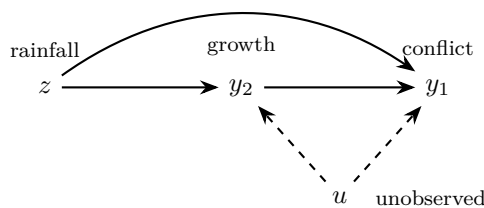


Figure 2: Sarsons's critique as a DAG. Same diagram as Figure 1, plus a direct $z \rightarrow y_1$ arrow that exclusion denies. Evidence from regions where the $z \rightarrow y_2$ channel is physically blocked (downstream of dams) supports the existence of this arrow.

The methodological lesson is general. Exclusion restrictions are sometimes partially testable: if the IV is claimed to operate through one channel, find a setting where the channel is shut and check whether the IV still moves the outcome. The post-2015 standard for IV papers in development economics has internalised this test.

4 Data

We use the replication panel distributed by MSS via the Harvard Dataverse (doi:10.7910/DVN/27324). It covers 41 sub-Saharan African countries over 1981–1999. After constructing one lag of GDP growth and dropping country-years with missing controls, the estimation sample is 702 country-year observations.

The outcome `any_prio` is a PRIO indicator equal to one if a country experiences at least 25 battle deaths in a year. The endogenous regressor `gdp_g` is annual per-capita GDP growth. The excluded instruments are current and lagged rainfall growth, `GPCP_g` and `GPCP_g_1`, constructed from the Global Precipitation Climatology Project. Controls follow MSS: initial log income, lagged Polity 2, ethnic and religious fractionalisation, an oil-exporter dummy, log mountainous terrain, and log lagged population. Only lagged Polity 2 is time-varying; the rest are absorbed by country fixed effects in our preferred specifications.

Table 1: Descriptive statistics, estimation sample ($N = 702$).

Variable	N	Mean	SD	Min	Max
<code>any_prio</code>	702	0.269	0.444	0.000	1.000
<code>gdp_g</code>	702	−0.006	0.066	−0.474	0.319
<code>GPCP_g</code>	702	0.015	0.210	−0.550	1.680
<code>y_0</code>	702	1.16	0.90	0.316	4.84
<code>polity2l</code>	702	−3.55	5.55	−10	9
<code>ethfrac</code>	702	0.655	0.237	0.036	0.925
<code>relfrac</code>	702	0.487	0.185	0.000	0.783
<code>Oil</code>	702	0.117	0.321	0	1
<code>lmtnest</code>	702	1.58	1.43	0	4.42
<code>lpopl1</code>	702	8.77	1.20	5.75	11.7

Three features of the panel matter for what follows. The unconditional conflict frequency is 27%. Average growth is slightly negative (−0.6% per annum). The panel is democracy-poor on average (mean Polity 2 of −3.5 on a scale of −10 to 10).

5 Methodology

6 Results and diagnostics

7 Discussion

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